

THE HITCHHIKER'S GUIDE TO MARKUP ESTIMATION: ASSESSING ESTIMATES FROM FINANCIAL DATA

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Macroeconomic outcomes depend on the distribution of markups across firms and over time, making firm-level markup estimates key for macroeconomic analysis. Methods to obtain these estimates require data on the prices that firms charge. Firm-level data with wide coverage, however, primarily come from financial statements, which lack information on prices. We use an analytical framework to show that trends in markups over time or the dispersion of markups across firms can still be well-measured with such data. Measuring the average level of the markup does require pricing data, and we propose a consistent estimator for such settings. We validate the analytical results using simulations of a quantitative macroeconomic model and offer supporting evidence from firm-level administrative production and pricing data. Our analysis supports the use of financial data to measure trends in aggregate markups.

KEYWORDS: Macroeconomics, production functions, markups, competition.

1. INTRODUCTION

THE MARKUP OF PRICES over marginal costs is a key variable in the macro-economy. Aggregate markups determine the labor and capital share in national income. Dispersion in markups across firms affects the efficiency of resource allocation. Variation in markups over the business cycle may explain real effects of nominal shocks. To understand macroeconomic outcomes, economists must therefore have a comprehensive picture of the distribution of firms' markups across the economy and over time. Yet neither prices nor marginal costs are observed in data sets at macroeconomists' disposal. This is because firm-level data with economy-wide coverage are nearly always derived from income statements and balance sheets, which at most have information on assets, revenue, and costs.

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A sprawling literature in macroeconomics and international trade has relied on markup estimates derived from such financial data, for example to quantify and test theories of imperfect competition à la [Atkeson and Burstein \(2008\)](#) and [Kimball \(1995\)](#).¹ A careful and quantitative assessment of the accuracy of firm-level markup estimates from financial data is, however, lacking.

This paper assesses the degree to which markups can be recovered from data on financial statements. We do so using an analytical framework, simulations of a quantitative macro model, and empirical analyses using French firm-level production and pricing data. We find that dispersion of markups across firms and trends over time can be well-estimated with financial data, provided that firms share a common production technology. This is the case if production is Cobb–Douglas and, more broadly, under empirically relevant conditions for more general production functions. Measuring average markups, however, requires data on prices, unless researchers employ methods that impose a demand system.

All parts of our analysis leverage the fact that, for cost-minimizing firms, markups equal the wedge between the elasticity of a firm's output with respect to a variable input—which firms set without adjustment costs—and that input's share in revenue ([De Loecker and Warzynski \(2012\)](#), [Hall \(1986, 1988\)](#)). This is the production approach to markup estimation. As inputs' revenue shares are observed in financial statements, measuring the output elasticity is a main challenge when estimating firm-level markups. This usually involves estimating a production function, as is done in the pioneering paper by [De Loecker and Warzynski \(2012\)](#).

The paper starts with an analytical framework, in which we characterize the biases that arise from a key issue when estimating markups using the production approach: financial statements only detail firm revenues, not the quantity that they produce. This affects estimation of the variable input's output elasticity: if firms are price setters, revenue is not proportional to output, and hence revenue cannot be used to consistently estimate the production function ([Klette and Griliches \(1996\)](#)). When revenue is used as a proxy for output, we show that the estimated output elasticities are biased by the correlation between firms' prices and the estimation's instruments, which often include last period's inputs. In simple models, that correlation is pinned down by an average of the price elasticity of demand. That average elasticity, in turn, determines average markups if firms are static profit maximizers. As a result, the estimated average of the markups is not informative about the true average. We show, however, that variation in markups across firms or over time can still be accurately measured, as long as firms have similar output elasticities. If firms differ in their output elasticities, revenue-based markup estimates correlate with true markups, and we show that the correlation approaches 1 if prices are orthogonal to the instruments.

These results appear at odds with the influential claim in [Bond, Hashemi, Kaplan, and Zoch \(2021\)](#) that revenue-based markups are uninformative about true markups. We ex-

¹This literature has been fueled by rising aggregate markup estimates over time (e.g., [De Loecker, Eeckhout, and Unger \(2020\)](#), [Diez, Fan, and Villegas-Sánchez \(2021\)](#)). These papers study the aggregate cost of markup ([Baqaee and Farhi \(2019\)](#), [Edmond, Midrigan, and Xu \(2023\)](#)), the role of markups in inequality ([Boar and Midrigan \(2025\)](#)), markup cyclicity ([Hong \(2017\)](#), [Burstein, Carvalho, and Grassi \(2025\)](#)), trade ([Edmond, Midrigan, and Xu \(2015\)](#), [Gaubert and Itskhoki \(2021\)](#)), the monetary policy transmission ([Baqaee, Farhi, and Sangani \(2024\)](#), [Chiavari, Morazzoni, and Smirnov \(2021\)](#), [Meier and Reinelt \(2022\)](#)), inflation dynamics ([Kouvavas, Osbat, Reinelt, and Vansteenkiste \(2021\)](#)), or price stickiness ([Wang and Werning \(2022\)](#), [Mongey \(2017\)](#)).

plain that their reasoning may hold on average, in the sense that the average revenue-based markup is usually not informative of the true average markup.²

The paper then validates these theoretical arguments through quantitative Monte Carlo simulations. The simulations enable us to scrutinize markup estimates from financial statements, where researchers lack data on prices, in a setting where true markups are known. We simulate a rich macroeconomic model of oligopolistic competition à la [Atkeson and Burstein \(2008\)](#) with endogenously heterogeneous markups. Firms differ in their fixed input and productivity, but share the same translog production function. For standard calibrations of the parameters and for various robustness calibrations, we find strong correlations between true markups and the various estimated markups. In a perfect scenario where one has data on the firms' quantity, markups are estimated with precision. In the practical scenario where researchers lack data on price and quantity, we still find a correlation of 0.94 between estimated and true markups. The high correlation is driven by the fact that in the full quantitative model, with various idiosyncratic and aggregate shocks, firms' prices and the instruments of the production function are close to orthogonal. Variation in markups—in the cross-section and over time—is also well-estimated, in line with the analytical results.

Finally, we compare revenue- and quantity-based markup estimates using administrative data on quantities and prices for French manufacturing firms. The data mostly validate the theoretical results. We find a 0.63 correlation between revenue and quantity markups, rising to 0.80 in first differences. These correlations are within sectors, consistent with the level of analysis in the theory and simulations. As in the simulations, we find that correlations between prices and the estimation instruments are low across all sectors. Binscatters show that revenue markups are good predictors of quantity markups across the domain. Regression coefficients that relate revenue and quantity markups to profits, labor- and market shares are also similar. Markup estimates do appear to be more dispersed when based on quantity, possibly due to several empirical complications: firms may not share identical production functions within sectors, for example, and functional forms for the production function and productivity process may be misspecified. For aggregate markups in France, we again find that the level is mismeasured when using revenue data. Trends over time, however, are well-estimated, and largely robust to different ways of aggregating markups.

We consider various extensions. If the variable input is subject to further input wedges, we show that markup estimates equal the product of markups and such wedges. The methods can also accommodate quality differences across firms.

Overall, we conclude that firm-level estimates of markups from the production approach are informative of true markups. However, our results do imply that researchers should give careful consideration to the suitability of their data for the question at hand. When interested in the level of markups, one generally needs quantity data. Revenue-based markups should not be used, for example, to discipline parameters that govern aggregate markups. When interested in variation of markups across firms or trends over time, revenue data will likely suffice.

²Bond, Hashemi, Kaplan, and Zoch (2021) write “This approach uses the revenue elasticity for a flexible input, in place of the output elasticity” and that the resultant markup is “(...) identically equal to one, and therefore contains no useful information about markups.” Their conclusion and Appendix B.2. do note that it may be possible to study variation in markups without estimating production functions (and thus without data on quantity), employing data on the variable input's revenue share and firms' overall input usage.

Related Literature. We contribute to the large and growing literature that uses firm-level markups to understand the macroeconomic implications of imperfect competition. This literature relies on firm-level markup estimates across the entire economy and on long time windows to quantify theoretical models. Recent examples include [Baqaee and Farhi \(2019\)](#) and [Edmond, Midrigan, and Xu \(2023\)](#), who study the cost of markup dispersion, [Boar and Midrigan \(2025\)](#), who study the role of markups in inequality, or [Hong \(2017\)](#) and [Burstein, Carvalho, and Grassi \(2025\)](#), who study markups over the business cycle.³ We show that markup estimates from revenue data can be used to calibrate parameters relating to markup dispersion or relative markups across firms, but not for parameters that govern the average markup.

Methodologically, our paper relates most closely to work that estimates markups using the production approach. This method originates from the sector-level estimator in [Hall \(1986, 1988\)](#), who derives that the markup is equal to the wedge between a variable input's output elasticity and the input's revenue share.⁴ In their seminal paper, [De Loecker and Warzynski \(2012\)](#) propose a generalized version of this estimator to measure firm-level markups, using production function estimation to obtain the output elasticity. If that elasticity were known, a variable input's revenue share suffices to measure markups, leaving the choice of a variable input as the sole challenge.⁵ In practice, output elasticities must be estimated—for which [Akerberg, Caves, and Frazer \(2015\)](#)'s method is typically used. This two-stage procedure first purges output of measurement error and transitory productivity shocks, after which the elasticities are identified using instruments.

Our paper is closely related to recent work that criticizes the use of these production function estimation techniques when estimating markups. [Bond, Hashemi, Kaplan, and Zoch \(2021\)](#) and [Doraszelski and Jaumandreu \(2021\)](#) point out that these techniques typically assume that firms are price takers—an unfortunate assumption when estimating markups. For price-taking firms, revenue is proportional to output, but price-setting firms must reduce prices when raising output. Revenue elasticities are thus generally lower than output elasticities (e.g., [Klette and Griliches \(1996\)](#)), causing an underestimation of markups ([De Loecker and Warzynski \(2012\)](#)). This does not mean quantity data are necessarily better, as quantities may be difficult to compare across firms due to product differentiation ([De Loecker \(2021\)](#)).

A particularly strong critique is found in [Bond, Hashemi, Kaplan, and Zoch \(2021\)](#), who claim there is no information about true markups in estimates that rely on revenue to estimate output elasticities. We explain that this is only correct if the production function estimation delivers the revenue elasticity, which is typically not the case. The average revenue-based markup is not informative of the true average, but variation in markups is well-measured. This matters, as the paucity of price data, as well as the limited number

³Footnote 1 provides additional examples of recent papers.

⁴A significant body of the industrial organization literature, alternatively, derives markups from firms' first-order condition for pricing after estimating a demand system. This "demand approach," which builds on [Berry, Levinsohn, and Pakes \(1995\)](#), enables markup estimation at the product level. Recent contributions include [AFSS+ \(2023\)](#), [Döpfer, MacKay, Miller, and Stiebale \(2022\)](#), [Miller, Osborne, Sheu, and Sileo \(2023\)](#), and [Grieco, Murry, and Yurukoglu \(2024\)](#). Macroeconomists often prefer the production approach, as it does not impose a demand system.

⁵This method, used in, for example, [Burstein, Carvalho, and Grassi \(2025\)](#), [Meier and Reinelt \(2022\)](#), [Calligaris, Criscuolo, and Marcolin \(2018\)](#), and [De Ridder \(2024\)](#), is a main alternative to cost shares estimators. A primary advantage is that it imposes little structure on the production function, which cost-share approaches do. One advantage of cost-share approaches is that they may be robust to factor-augmenting productivity ([Raval \(2023a,b\)](#)).

of settings in which firms' quantities are comparable, mean that [Bond, Hashemi, Kaplan, and Zoch](#)'s claim seriously limited the scope for future analysis of markups.⁶

[Bond, Hashemi, Kaplan, and Zoch \(2021\)](#) do note that there are applications in which it is possible to study variation in markups across firms without estimating a production function. This uses the fact that output elasticities are a function of inputs, and inputs can be added as controls in regressions. Our findings are complementary, as we conclude that variation in revenue-based markup estimates can be relied upon even in settings where it is not feasible to control for the output elasticities. Our results furthermore preserve the credibility of important prior contributions in macroeconomics and international trade that use revenue-based markup estimates to study variation in markups across firms and over time.⁷

A final issue is that production function estimates may be biased using [Akerberg, Caves, and Frazer \(2015\)](#)'s procedure if firms are price setters, even with quantity data.⁸ [Doraszelski and Jaumandreu \(2019\)](#) and [Brand \(2019\)](#) point out that an identifying assumption in [Akerberg, Caves, and Frazer \(2015\)](#) is that demand faced by firms is not affected by unobservables other than productivity, which may not be true under oligopolistic competition. We propose a change to the procedure's first stage to address this.⁹ A new paper by [Akerberg and De Loecker \(2024\)](#) provides a detailed review of production function estimation under imperfect competition.

2. ANALYTICAL FRAMEWORK

In this analytical section, we first discuss how [Hall \(1986, 1988\)](#) and [De Loecker and Warzynski \(2012\)](#) relate markups to production functions. We follow the latter's more

⁶Even when quantity data are available, such data can be misleading in production function estimation under certain conditions. For instance, if firms sell a homogeneous product in different package sizes at the same per-volume price, quantity data are less informative about output than revenue.

⁷[De Loecker and Warzynski \(2012\)](#) acknowledge that data on comparable quantities are needed to estimate average output elasticities and markups with precision. Our paper provides closed-form expressions for biases that arise in the absence of such data, as well as Monte Carlo and empirical evidence on the degree to which markup dispersion across firms and over time remains well-measured. [De Loecker \(2011\)](#) notes that specifying a demand system can also address biases in production function estimation due to missing firm-level price data. This solution is not suitable in a macroeconomic context as it reduces the generality of the conditions under which the markup estimates are obtained.

⁸[Akerberg, Caves, and Frazer \(2015\)](#)'s method is an example of the control function approach, developed to estimate production functions and productivity under perfect competition. The control function is a component of the purging regression, where measurement error is identified as the residual from regressing output on observables. To ensure productivity is not inadvertently purged, the observables must be an invertible function of productivity. The approach was pioneered by [Olley and Pakes \(1996\)](#) using investment to control for productivity. [Levinsohn and Petrin \(2003\)](#) modify this approach by proposing static controls (e.g., a variable input) to address the practical issue that observed investment is lumpy, and thus not an invertible function of productivity. [Akerberg, Caves, and Frazer \(2015\)](#) amend the procedure by identifying all coefficients in the second stage rather than the first, addressing collinearity between productivity and variable inputs. We elaborate in Section 2. A detailed discussion of the history of production function estimation, including a summary of practical issues with early methods, can be found in [Akerberg, Caves, and Frazer \(2015\)](#). [De Loecker and Syverson \(2022\)](#) provide a careful contemporaneous discussion, including a review of differences between control function approaches and dynamic panel approaches (e.g., [Blundell and Bond \(2000\)](#)) for production function estimation.

⁹We do need sufficient variation in input prices so that inputs and productivity are not collinear ([Blundell and Bond \(2000\)](#), [Gandhi, Navarro, and Rivers \(2020\)](#)). Further discussion on markup estimation with accounting data is found in [Traina \(2018\)](#), [Morlacco \(2019\)](#), [Basu \(2019\)](#), [Syverson \(2019\)](#), [Yeh, Macaluso, and Hershbein \(2022\)](#).

general derivation, which applies at the firm level. We then outline challenges to production function estimation with commonly available data sets if firms have market power. Finally, we derive expressions for the biases that arise when revenue is used to approximate output, or in case of measurement error.

2.1. From Markups to Production Functions

The seminal papers by Hall (1986, 1988) and De Loecker and Warzynski (2012) propose a method to estimate markups that relates markups to the production function. The idea is that markups can be inferred from the wedge between the output elasticity of a variable input and that input's share in revenue. An input is variable if firms choose its use every period to minimize costs, without intertemporal considerations or adjustment costs, taking the price of this input as given.

Formally, the output Y_{it} for firm i at time t is given by the production function $Y_{it} = Y(V_{it}, \mathbf{K}_{it}, \Omega_{it})$, where V_{it} is the variable input, purchased at price W_t . The vector $\mathbf{K}_{it}$ contains all other inputs, while Ω_{it} represents productivity. The first-order condition for the cost-minimizing firm with respect to V_{it} is given by $\frac{1}{\eta_{it}} = \frac{\partial Y_{it}}{\partial V_{it}} \frac{1}{W_t}$, where η_{it} is the Lagrange multiplier of the production function constraint and is equal to marginal costs. Multiplying both sides by price P_{it} yields the markup expression that we use throughout the analysis:

$$\mu_{it} = \alpha_{it} \frac{P_{it} Y_{it}}{W_t V_{it}}, \quad (1)$$

where $\mu_{it} \equiv P_{it}/\eta_{it}$ is the markup and $\alpha_{it} = \frac{\partial Y_{it}}{\partial V_{it}} \frac{V_{it}}{Y_{it}}$ is the output elasticity of V_{it} .

The expression yields the familiar result that an input's output elasticity equals its revenue share if markups are 1, while revenue shares fall short of the output elasticity when markups exceed 1. It follows that to estimate markups, researchers need data on revenue and input spending from the income statement, as well as an estimate of α_{it} , the output elasticity of V_{it} . Estimating this elasticity under imperfect competition is thus a primary challenge in markup estimation.

It is worth noting that for inputs not conforming to the assumptions placed on V_{it} (i.e., adjustment costs for capital or imperfections in the labor market), that input's equivalent of equation (1) can be used to calculate the overall input wedge, as defined by Hsieh and Klenow (2009). We discuss this further in Section 2.4.

2.2. Estimating Markups With Revenue Data

We next introduce an analytical framework to analyze whether markups can be measured along (1) when estimating α_{it} using commonly available data sets. As these markups inform macroeconomic and international trade theories, it is crucial for data to have economy-wide coverage, preferably over long time horizons. Data sets meeting this bar are usually based on financial statements, where revenue is the sole measure of output. As firms' decisions influence prices under imperfect competition, however, revenue may poorly approximate output.

2.2.1. When Do Revenue-Based Markups Measure True Markups?

We first examine how accurately revenue-based markup estimates measure true markups, showing that their correlation depends on the estimated output elasticity of

the variable input. If the estimated elasticity matches the true output elasticity, true markups are recovered; if it matches the elasticity of revenue with respect to the variable input—the *revenue elasticity*—markup estimates are orthogonal to true markups; if it falls between these elasticities, markup estimates positively correlate with true markups.

To derive this, we introduce a simple demand system. Firms face a price-elasticity of demand $\frac{dy_{it}}{dp_{it}} = -1/\varepsilon_{it}$, where lowercase letters denote the log deviation from a sample mean. The demand elasticity can be heterogeneous across firms and over time. Firms that maximize profits period-by-period in the face of this demand will charge a markup $\mu_{it} = (1 - \varepsilon_{it})^{-1}$. This is the standard inverse elasticity rule that describes how firms set prices in static models of oligopolistic or monopolistic competition. The demand system gives us the elasticity of revenue $R_{it} = P_{it} Y_{it}$ with respect to V_{it} as $\frac{dr_{it}}{dv_{it}} = \frac{dp_{it}}{dy_{it}} \frac{dy_{it}}{dv_{it}} + \frac{dy_{it}}{dv_{it}} = (1 - \varepsilon_{it})\alpha_{it}$.

With true markups delivered by the demand system, we can now derive the correlation between true markups and revenue-based markup estimates. These estimates use equation (1), where the output elasticity α_{it} is replaced by the estimated elasticity on revenue data, $\hat{\alpha}_{it}$. The literature has developed techniques to estimate parameters of a production function with firm data. When such techniques use revenue in place of quantity data, the production function parameters (and hence $\hat{\alpha}_{it}$) are biased (Klette and Griliches (1996)). The bias is a function of the joint distribution of inputs and of the elasticities of both demand and output, as we show in Section 2.2.2. Hence, revenue-based markup estimates read as

$$\hat{\mu}_{it}^R \equiv \hat{\alpha}_{it} \frac{P_{it} Y_{it}}{W_{it} V_{it}} = \frac{\hat{\alpha}_{it}}{\alpha_{it}} \alpha_{it} \frac{P_{it} Y_{it}}{W_{it} V_{it}} = \frac{\hat{\alpha}_{it}}{\alpha_{it}(1 - \varepsilon_{it})} = \frac{\hat{\alpha}_{it}}{\alpha_{it}} \mu_{it}. \quad (2)$$

This elucidates the relationship between true and estimated markups, and true and estimated output elasticities. The final equality confirms that if the estimated elasticity $\hat{\alpha}_{it}$ equals the true elasticity, revenue-based markups recover true markups. The penultimate equality shows that the revenue-based estimates $\hat{\mu}_{it}^R$ can correlate with true markups μ_{it} as long as $\hat{\alpha}_{it}$ is different from the revenue elasticity $\alpha_{it}(1 - \varepsilon_{it})$. In such cases, $\text{Cov}[\log \mu_{it}, \log \hat{\mu}_{it}^R] \neq 0$. However, if markups are computed using the true revenue elasticity $\alpha_{it}(1 - \varepsilon_{it})$ in place of α_{it} , we have

$$\hat{\mu}_{it}^{\text{RE}} \equiv (1 - \varepsilon_{it})\alpha_{it} \frac{P_{it} Y_{it}}{W_{it} V_{it}} = (1 - \varepsilon_{it})\mu_{it} = \frac{1 - \varepsilon_{it}}{1 - \varepsilon_{it}} = 1.$$

That is, the markup estimates would be identically equal to 1 and would be orthogonal to true markups: $\text{Cov}[\log \mu_{it}, \log \hat{\mu}_{it}^{\text{RE}}] = 0$, as in Bond, Hashemi, Kaplan, and Zoch (2021).

2.2.2. Estimating the Output Elasticity With Revenue Data

We next show that, in practice, estimating a parametric production function using revenue instead of quantity yields neither the revenue nor the output elasticity, as long as firms have heterogeneous markups. Rather, the estimate of the output elasticity $\hat{\alpha}_{it}$ that one obtains when using revenue instead of quantity data suffers from an omitted variable bias. When quantity is used to estimate the production function, the output elasticity can be consistently estimated using standard tools, as we discuss in the Supplemental Material, De Ridder, Grassi, and Morzenti (2026).¹⁰ Below, use a simple analytical framework to derive the omitted variable bias that arises when using revenue instead.

¹⁰We show that an IV-GMM estimator is consistent and recovers the output elasticity for the simple analytical framework (Supplemental Appendix A.1) and in more general cases (Supplemental Appendix A.4).

Consider a set of firms in a single sector. Output Y_{it} is log-linear in a single variable input V_{it} , while productivity is identically and independently distributed across firms and time. Productivity is unobserved by the researcher but observed by the firm. Firms set V_{it} and share the same Cobb–Douglas production function

$$y_{it} = \alpha v_{it} + \omega_{it}, \quad (3)$$

where lower caps denote log deviations from the mean, and where α is the true output elasticity of v_{it} to be estimated.¹¹ This simple environment allows us to keep the argument as transparent as possible and to derive clear closed-form solutions. Despite the simplicity, the intuitions extend to more general models that are standard in the literature. In Section 2.2.4, for example, we study the case of a translog production function where the output elasticity is heterogeneous across firms. Supplemental Appendix A.4 furthermore extends the results by allowing for multiple inputs (A.4.2), persistence in productivity (A.4.3), and all of these together (A.4.4).¹²

Turning to the estimation of α , a least-squares regression of input v_{it} on output y_{it} will be biased even if output is observed. This is because productivity affects a firm's input choices and is unobserved to the econometrician, and is thus in the residual. The production function estimation literature, such as [Blundell and Bond \(2000\)](#) or [Akerberg, Caves, and Frazer \(2015\)](#), identifies α by instrumenting v_{it} by its lag v_{it-1} . In our setup, since productivity is i.i.d., v_{it-1} is not in the same information set and thus is orthogonal to ω_{it} .¹³ The instrument therefore satisfies the exclusion restriction. For instrument relevance, we also need v_{it} to be persistent over time. This might arise through persistence in input price W_t . [Gandhi, Navarro, and Rivers \(2020\)](#) note that under perfect competition, this is the sole driver of persistence, so that long time samples are required for identification. We show that under imperfect competition (the natural setting when estimating markups), it is easier to obtain persistence in v_{it} , as a firm's persistent set of competitors affects its demand for inputs (Supplemental Appendix A.1). Hence, the output elasticity α can be identified.

When estimating the production function with revenue data, one obtains a biased estimate of α . To show this, let us construct an instrumental variable estimator based on the generalized method of moments (IV-GMM) when revenue is used in place of quantity. We focus on infinite samples to abstract from finite-sample variation. Hence, $\mathbb{E}[x_{it}]$ denotes the limit in probability of the sample average of a variable x_{it} as the sample size goes to infinity.¹⁴ We thus focus on consistency of the estimator. Revenue is quantity times price, such that $r_{it} = y_{it} + p_{it}$ is revenue in log deviations from the mean. Furthermore,

¹¹To be precise, $x_{it} = \log X_{it} - \mathbb{E}[\log X_{it}]$, where $\mathbb{E}[\log X_{it}]$ is the limit of the sample average. This normalization gets rid of constants in the production function and ensures that ω_{it} is mean zero.

¹²Supplemental Appendix A.4.3 notes that if productivity has persistence ρ , identification may only hold locally. In particular, there are exactly two solutions to the IV-GMM estimator. One matches the true parameters, while the second is a biased estimate of the true parameters. This is in line with recent work by [Akerberg, Frazer, Luo, and Su \(2023\)](#), who show independently that the two-stage estimator may have two solutions, rendering standard numerical solvers unstable. However, in our framework, we show that if $\text{Var}[v_{it-1}]$ is large compared to $\text{Var}[\omega_{it-1}]$ and $\text{Var}[v_{it} - \rho v_{it-1}]$, then there exists a unique solution for $\hat{\alpha}$ and $\hat{\rho}$. Hence, if there is enough variation in the data, the production function is globally identified.

¹³Supplemental Appendix A.4.3 notes that when productivity is persistent, for example, when it is AR(1), v_{it-1} is still a valid instrument if the moment condition is for v_{it-1} to be orthogonal to innovations of productivity. A similar argument applies there: v_{it-1} is not in the same information set as the time t innovations.

¹⁴By the weak law of large numbers, under independence of the x_{it} , $\mathbb{E}[x_{it}] \equiv \text{plim } \frac{1}{N} \sum_{it} x_{it}$ also denotes the expected value of x_{it} . Supplemental Appendix A.2 derives the estimator for a finite size sample of firms.

inserting production function (3) for y_{it} yields $r_{it} = y_{it} + p_{it} = \alpha v_{it} + \omega_{it} + p_{it}$, where α remains the parameter of interest. If a researcher were to use an IV-GMM estimator that is consistent for quantity, but uses revenue as the dependent variable instead, that estimator would be defined as follows:

DEFINITION 1—Revenue IV-GMM Estimator: The estimator is $\hat{\alpha} \in \mathbb{R}$ such that moment $\mathbb{E}[\widehat{\text{tfpr}}_{it} v_{it-1}]$ is zero, where $\widehat{\text{tfpr}}_{it} = p_{it} + y_{it} - \hat{\alpha} v_{it} = (\alpha - \hat{\alpha}) v_{it} + p_{it} + \omega_{it}$.

Without loss of generality, we treat $\hat{\alpha}$ as a non-random real number. Formally, $\hat{\alpha}$ is a random variable which is almost surely equal to a constant. Solving for $\hat{\alpha}$ such that $0 = \mathbb{E}[\widehat{\text{tfpr}}_{it} v_{it-1}] = (\alpha - \hat{\alpha}) \mathbb{E}[v_{it} v_{it-1}] + \mathbb{E}[p_{it} v_{it-1}]$ yields a unique solution as long as the lagged variable input is a relevant instrument (i.e., $\mathbb{E}[v_{it} v_{it-1}] \neq 0$):

$$\hat{\alpha} = \alpha + \frac{\mathbb{E}[p_{it} v_{it-1}]}{\mathbb{E}[v_{it} v_{it-1}]} \quad (4)$$

The estimator is clearly not consistent if prices and lagged variable inputs covary, that is, $\mathbb{E}[p_{it} v_{it-1}] \neq 0$. Using revenue to measure output creates an omitted variable bias: the revenue-production function has prices in the residual, as first pointed out by Klette and Griliches (1996) and discussed in De Loecker, Goldberg, Khandelwal, and Pavcnik (2016).

Under perfect competition, the correlation between price and lagged inputs is zero, as firms are atomistic price takers. Under imperfect competition, it is probable that p_{it} will correlate with lagged variable inputs, such that $\mathbb{E}[p_{it} v_{it-1}]$ differs from zero. Note that there are no model-free constraints on either the size or sign of the covariance. If firms face persistent aggregate demand shocks and decreasing returns to scale, for example, positive shocks drive up marginal costs and prices, causing a *positive* correlation between prices and lagged variable inputs. Conversely, firms with downward-sloping demand curves reduce prices to sell additional output, causing a *negative* correlation. The estimates of α can therefore be smaller, larger, or equal to the true output elasticity. Equally, the ensuing markup estimates may overstate, understate, or equal true markups.

2.2.3. Revenue-Based Markup Estimates

We next show that, even when biased, revenue-based markup estimates are still informative about true markups. The bias in the estimated elasticity in equation (4) is, in part, determined by the demand system—and so to show this, we reintroduce a demand side to our baseline framework. We assume a general invertible demand system, where a firm's demand depends on prices of all firms. Formally, the vector of quantities produced by all firms, $\{Y_{it}\}$, is a function of the price vector $\{P_{it}\}$ such that $\{Y_{it}\} = D_t(\{P_{it}\})$. A log-linear approximation yields

$$p_{it} = - \sum_j \varepsilon_{ijt} y_{jt}, \quad (5)$$

where ε_{ijt} is the cross-elasticity of firm i 's price to firm j 's quantity. For now, we abstract from aggregate shocks that alter price-quantity relationships across periods, and hence focus on the bias caused by downward-sloping demand curves.

With this demand system, the revenue elasticity of the variable input, taking other firms' output as given, is $\frac{dr_{it}}{dv_{it}} = \alpha(1 - \varepsilon_{iit})$. The variable input v_{it} can be used to compute markups along equation (1); as firms share a common output elasticity α , true markups equal $\mu_{it} =$

$\alpha(P_{it}Y_{it})/(W_tV_{it})$. The estimated elasticity, substituting demand system (5) into equation (4) and using production function (3), is given by

$$\hat{\alpha} = \alpha \left(1 - \sum_j \frac{\mathbb{E} \left[\varepsilon_{ijt} \left(v_{jt} + \frac{\omega_{jt}}{\alpha} \right) v_{it-1} \right]}{\mathbb{E}[v_{it}v_{it-1}]} \right).$$

The ratio of the true output elasticity and the revenue-based estimated elasticity is equal to 1 minus the weighted average of inverse demand elasticities and cross-elasticities among the firms sharing the same production function. Importantly, the estimated elasticity $\hat{\alpha}$ differs, generally, from the revenue elasticity $\alpha(1 - \varepsilon_{iit})$. This implies that the revenue markup is different from 1, as in equation (2). Indeed, firm-level markup estimates based on revenue data, $\hat{\mu}_{it}^R = \hat{\alpha} \frac{P_{it}Y_{it}}{W_tV_{it}}$, are

$$\hat{\mu}_{it}^R = \mu_{it} \left(1 - \sum_j \frac{\mathbb{E} \left[\varepsilon_{ijt} \left(v_{jt} + \frac{\omega_{jt}}{\alpha} \right) v_{it-1} \right]}{\mathbb{E}[v_{it}v_{it-1}]} \right).$$

This shows that revenue-based markup estimates are equal to true markups up to a constant, as the second term in parentheses is not firm-specific (the $\mathbb{E}$ takes an average over i). The true and estimated revenue log markups have equal variances and the correlation between the estimates and true markups is 1.

The result that revenue and quantity markups perfectly correlate depends on the Cobb–Douglas assumption that output elasticity is constant. The bias is a constant, and thus does not cancel out variation in markups. In Section 2.2.4, we discuss the case of variable output elasticities and show that the core insights remain: variation in the bias does not cancel out variation in markups.

Case I: Heterogeneous Demand Elasticities. The result that revenue-based estimates of the markup perfectly covary is only useful if markups are variable across firms. In our simple demand system, in which firms set prices to maximize contemporaneous profits, firms will have heterogeneous markups if they are subject to heterogeneous price elasticities of demand. For this case, it is straightforward to derive that all dispersion in the markup is preserved, but mean markups may be sufficiently biased such that *no information* about the true average remains.

To see this, start from the demand system in equation (5), with the additional assumption that for all pairs of firms i, j with $i \neq j$, $\varepsilon_{ijt} = 0$ while $\varepsilon_{iit} \neq 0$ and $\varepsilon_{iit} \neq \varepsilon_{jjt}$. Thus, demand is determined by firms' own supply, and firms face heterogeneous demand elasticities. Formally, the demand system is such that $p_{it} = -\varepsilon_{it}y_{it}$, where, with some abuse of notation, we denote $\varepsilon_{it} \equiv \varepsilon_{iit}$ as the own inverse price elasticity. When firms maximize profits, they charge a markup $\mu_{it} = (1 - \varepsilon_{it})^{-1}$.

Under these assumptions, revenue IV-GMM estimators give the *average* revenue elasticity among firms sharing a production function. This differs from each firm's own rev-

enue elasticity, because firms have different demand elasticities:¹⁵

$$\hat{\alpha} = \mathbb{E} \left[\alpha(1 - \varepsilon_{it}) \frac{v_{it} v_{it-1}}{\mathbb{E}[v_{it} v_{it-1}]} \right] \neq \frac{\partial r_{it}}{\partial v_{it}} = \alpha(1 - \varepsilon_{it}). \quad (6)$$

Turning to the resultant markup estimates along equation (1) and for markups $\mu_{it} = (1 - \varepsilon_{it})^{-1}$ that maximize profits, we get

$$\hat{\mu}_{it}^R \equiv \hat{\alpha} \frac{P_{it} Y_{it}}{W_t V_{it}} = \mathbb{E} \left[\mu_{it}^{-1} \frac{v_{it} v_{it-1}}{\mathbb{E}[v_{it} v_{it-1}]} \right] \mu_{it}. \quad (7)$$

As in the previous case, the revenue-based markup estimates equal the true markups up to a constant. For our simple demand system, this constant is equal to the weighted average of inverse markup among firms sharing the same production function. Given the assumption that two firms i and j have different markups, the estimated revenue markup $\hat{\mu}_{it}^R$ is different from 1. However, the average of the estimated revenue markup is not informative about the average true markup. Indeed, the average estimated revenue markup can be written as

$$\mathbb{E}[\hat{\mu}_{it}^R] = \mathbb{E} \left[\mu_{it}^{-1} \frac{v_{it} v_{it-1}}{\mathbb{E}[v_{it} v_{it-1}]} \right] \mathbb{E}[\mu_{it}],$$

which equals 1 up to a Jensen's inequality. Revenue markups thus carry no information about the true average in this demand system. We conclude that using revenue data instead of quantity data does not allow us to recover the level of markups but allows us to recover variation in markups, if variation exists.

Case II: Homogeneous Demand Elasticities. There is one case where revenue-based markup estimates do not contain any useful information about true markups. This is when firms compete monopolistically and have identical price-elasticities of demand such that $p_{it} = -\gamma y_{it}$. This assumption is satisfied by constant elasticity of substitution (CES) preferences with atomistic firms if the aggregate price index is fixed. In that case, the revenue estimator equals the revenue elasticity $\hat{\alpha} = \alpha(1 - \gamma) = \frac{\partial y_{it}}{\partial v_{it}} (1 + \frac{\partial p_{it}}{\partial y_{it}}) = \frac{\partial r_{it}}{\partial v_{it}}$. Both the revenue elasticity and the true markup are equal across firms, where the latter is equal to $(1 - \gamma)^{-1}$. It follows that revenue-based markup estimates are identically equal to 1, $\hat{\mu}_{it}^R = (1 - \gamma)^{-1} (1 - \gamma) = 1$, as in Bond, Hashemi, Kaplan, and Zoch (2021). If markups are identical across firms, revenue markups thus do not yield any information on true markups.

2.2.4. Beyond Constant Output Elasticities

The analysis thus far assumes that firms have equal output elasticities. We next study the more general case where the output elasticity depends on firms' input choices, by assuming that the production function is translog. The translog production function nests Cobb–Douglas and is a second-order approximation of any production function. In our one-input environment, the function is given by

$$y_{it} = \alpha v_{it} + \beta v_{it}^2 + \omega_{it}, \quad \text{so that } \alpha_{it} \equiv \frac{\partial y_{it}}{\partial v_{it}} = \alpha + 2\beta v_{it}. \quad (8)$$

¹⁵This assumes that $\mathbb{E}[\varepsilon_{it} \omega_{it} v_{it-1}] = 0$. This assumption is satisfied (for example) when, conditional on v_{it-1} , productivity ω_{it} and elasticity ε_{it} are orthogonal or, alternatively, when conditional on ε_{it} , ω_{it} and v_{it-1} are orthogonal. We make this assumption merely to clarify the argument.

As for the case where firms have equal output elasticities, the bias from using revenue instead of quantity depends on the correlation between prices and the instruments in the production function estimation, which are typically the lags of v_{it} and v_{it}^2 . As we derive in Supplemental Appendix A.5, the revenue IV-GMM estimator gives

$$\begin{pmatrix} \hat{\alpha} \\ \hat{\beta} \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} + V^{-1} \begin{pmatrix} \mathbb{E}[p_{it}v_{it-1}] \\ \mathbb{E}[p_{it}v_{it-1}^2] \end{pmatrix}, \quad \text{with } V = \begin{pmatrix} \mathbb{E}[v_{it}v_{it-1}] & \mathbb{E}[v_{it}^2v_{it-1}] \\ \mathbb{E}[v_{it}v_{it-1}^2] & \mathbb{E}[v_{it}^2v_{it-1}^2] \end{pmatrix}.$$

If prices are orthogonal to the instruments, the second term disappears and the estimates are consistent. If either the lagged variable input or its square is correlated with prices, both parameters are biased, as is the estimated elasticity $\hat{\alpha}_{it} = \hat{\alpha} + 2\beta v_{it}$. Inserting this into equation (1) gives the revenue markup:

$$\hat{\mu}_{it}^R = (1 + f(v_{it}))\mu_{it}, \quad \text{where } f(v_{it}) = \frac{(\hat{\alpha} - \alpha) + 2(\hat{\beta} - \beta)v_{it}}{\alpha + 2\beta v_{it}},$$

which equals true markups multiplied by a function of firms' own input usage. If prices are orthogonal to the instruments, markups are accurately estimated.

As for the case of constant output elasticities, we can impose a demand system to shed more light on what might drive a correlation between prices and instruments, and how that affects both the level and dispersion of markup estimates.

Case I: Heterogeneous Demand Elasticities. If firms have heterogeneous demand elasticities and face the previously specified demand system, $p_{it} = -\varepsilon_{it}y_{it}$, variation in markups can be accurately measured if both v_{it} and the instruments are orthogonal to ε_{it} . In that case, we get $(\hat{\alpha}, \hat{\beta}) = (1 - \mathbb{E}[\varepsilon_{it}])(\alpha, \beta)$. The markup estimates equal $\mu_{it}^R = (1 - \mathbb{E}[\varepsilon_{it}])\mu_{it}$ and have a correlation of 1 with true markups. Derivations are provided in Supplemental Appendix A.5. The average markup will be uninformative of the true average, however, as in the case of constant output elasticities.

We should note, however, that v_{it} is not necessarily orthogonal to the demand elasticity ε_{it} . In macroeconomic models such as Atkeson and Burstein (2008) and Kimball (1995), for example, large firms have lower demand elasticities. This means that the correlation between revenue-based markup estimates and true markups may be below 1. Even in those settings, however, the prices and instruments may be close to orthogonal because prices are determined by a combination of returns to scale, a firm's direct competitors, aggregate demand conditions, and factor prices. The extent to which revenue markups and true markups correlate is thus a quantitative question that we answer with simulations in Section 4 and empirically in Section 5. In both, we find high correlations between revenue and true markups, and negligible correlations between prices and instruments.

Case II: Homogeneous Demand Elasticities. Revenue markups are uninformative of true markups, $\hat{\mu}_{it}^R = 1$, if firms have homogeneous demand elasticities (i.e., $p_{it} = -\gamma y_{it}$). The logic extends from the constant output elasticity case: the average output elasticity from revenue data is biased by the demand elasticity, which also determines the markup. Supplemental Appendix A.5 provides a formal derivation.

2.2.5. Analyzing Markup Variation Without Estimating Production Functions

We should note that there are research designs in which variation in markups can be analyzed without estimating the production function, rendering a lack of quantity data

inconsequential (De Loecker and Warzynski (2012), Bond, Hashemi, Kaplan, and Zoch (2021)). This still leverages the insight from the production approach to markup estimation that markups are given by the product of the inverse revenue share of a variable input, $P_{it} Y_{it} / (W_t V_{it})$, and that input's output elasticity, α_{it} . If the output elasticity is some function of observables $\mathbf{h}_{it}$, such that $\alpha_{it} = \alpha(\mathbf{h}_{it})$, the output elasticity can be controlled for and variation in the input's revenue share can be used to assess variation in markups. For example, if markups are such that $\ln \mu_{it} = \gamma + \beta X_{it} + \epsilon_{it}$ and β is of interest, one can substitute out the markup and run

$$\ln P_{it} Y_{it} / (W_t V_{it}) = -\ln \alpha(\mathbf{h}_{it}) + \gamma + \beta X_{it} + \epsilon_{it}.$$

This is feasible, as for many production functions (including translog), the output elasticity is a function of inputs. As $P_{it} Y_{it} / (W_t V_{it})$ is observed in the data, adding controls for inputs in a sufficiently flexible functional form enables one to estimate β without knowledge of the production function. This does have limitations. It is infeasible to control for input usage, for example, if one is interested in the relationship between markups and firm size. The same holds for other analyses that macroeconomists often perform with markups, such as measuring misallocation, or within-between firm decompositions of markup trends over time.

2.3. Markups, Productivity, and Measurement Errors

After studying how the use of revenue instead of quantity data affects the estimation of markups, we now focus on the case where output data are available but imperfectly measured. Specifically, we now assume that output is observed as $\tilde{y}_{it} = \alpha v_{it} + \omega_{it} + \eta_{it}$, where $\tilde{y}_{it}$ is observed quantity (while y_{it} is true quantity), ω_{it} is the productivity observed by the firm, and η_{it} is measurement error and white noise productivity shocks that firms observe after making production decisions.

Prior work pays specific attention to η_{it} , for three reasons. First, observed output can contain significant measurement error. In our empirics, for example, we measure output by dividing revenue by unit values, which are in turn obtained from surveys. Second, the presence of η_{it} impedes estimating true productivity ω_{it} : even if the production function parameters are known, one can only recover productivity with measurement error, $\omega_{it} + \eta_{it}$. Third, measurement errors complicate production function estimation when ω_{it} follows a non-linear process.

Below, we explain that one can either estimate the production function with greater standard errors, or purge measurement error in a first-stage regression.

Retaining Measurement Errors. In the presence of measurement errors η_{it} , one can still use the standard IV-GMM estimator to estimate the production function (Blundell and Bond (2000)). This is the procedure proposed by Doraszelski and Jaumandreu (2019, 2021). In Supplemental Appendix A.3, we show that, in our simple framework, the estimator remains consistent. Note, however, that for a finite-sized sample, the estimator's asymptotic variance is proportional to $\mathbb{E}[\omega_{it}^2] + \mathbb{E}[\eta_{it}^2]$. This means that the estimator's variance increases in measurement errors' variance.

The main drawback is that this procedure cannot recover productivity. To see this, note that productivity is measured as the difference between output and the product of all inputs and their estimated output elasticities. For our framework, this is $\tilde{y}_{it} - \alpha v_{it} = \omega_{it} + \eta_{it}$. This correlates with true productivity, but the correlation fades as the ratio of variance of measurement error to productivity increases.

In Supplemental Appendix A.3, we furthermore discuss that measurement error can also impede consistency of the IV-GMM estimator if ω_{it} is persistent with non-linear autoregressive terms (Bond, Hashemi, Kaplan, and Zoch (2021), Akerberg and De Loecker (2024)). In our quantitative analysis, we further explore the impact of abstracting from measurement errors in estimating the production function and markups.

Purging Measurement Error From Quantity. The combination of the loss of direct estimates for true productivity ω_{it} , higher standard errors, and stringent assumptions on the dynamic process of ω_{it} form a case to purge observed output from measurement error. Akerberg, Caves, and Frazer (2015) do so in a first-stage purging regression for the case of perfect competition. We propose a procedure that—deviating minimally from theirs—can accomplish purging under imperfect competition.

The purging regression aims to separate η_{it} and ω_{it} , using the fact that firms only observe ω_{it} when deciding the quantity of inputs that they wish to deploy. The idea is that the demand for the variable input can therefore be expressed as a function of productivity: $v_{it} = v(\omega_{it}, \Xi_{it})$, where Ξ_{it} is a vector of all variables that determine v_{it} other than productivity. Under the assumption that v_{it} rises monotonically in ω_{it} , the demand function can be inverted, such that $\omega_{it} = v^{-1}(v_{it}, \Xi_{it})$. This function is often called the control function, as in Olley and Pakes (1996), Levinsohn and Petrin (2003), or Akerberg, Caves, and Frazer (2015). Previous literature on the control function approach is summarized in footnote 8. In our framework, the observed output can therefore be written as $\tilde{y}_{it} = \alpha v_{it} + v^{-1}(v_{it}, \Xi_{it}) + \eta_{it}$. The fitted values of a non-parametric regression of $\tilde{y}_{it}$ on v_{it} and Ξ_{it} therefore identify η_{it} , as long as Ξ_{it} contains all variables that determine the demand for v_{it} .¹⁶ The fitted values can then be used to construct moment $\mathbb{E}[\hat{\omega}_{it} v_{it-1}]$, a function of $\hat{\alpha}$. The $\hat{\alpha}$ that sets this moment to zero is a consistent estimator of the true α .

What variables are included in Ξ_{it} under imperfect competition? As noted by De Loecker and Warzynski (2012) and De Loecker and Syverson (2022), the determinants of v_{it} depend on the setting that is considered, including the competition that firms face. In our framework, one can readily find the variables in Ξ_{it} from the firm's cost-minimization problem. Inverting the demand function gives $\omega_{it} = (1 - \alpha)v_{it} - mc_{it} + w_t$. It follows that factor prices and (controls for) log marginal costs should be included in Ξ_{it} . Given that marginal costs can be expressed in terms of prices and markups, observed output can be written as¹⁷

$$\tilde{y}_{it} = v_{it} - p_{it} + \log \mu_{it} + w_t + \eta_{it}. \quad (9)$$

To purge for measurement error, it would therefore suffice to regress observed output on the variable input, prices, markups, and time-fixed effects for w_t .¹⁸ Under perfect competition, firms are price takers and have log markups of 0. Hence, a first-stage regression of output on v_{it} and time-fixed effects is sufficient to purge for measurement error.¹⁹ Under imperfect competition, this is not sufficient, as firms have heterogeneous markups.

¹⁶Our vector Ξ_{it} corresponds to the vector z in De Loecker and Warzynski (2012).

¹⁷Note that the expression of the marginal cost $MC_{it} = P_{it}/\mu_{it}$ in log deviation from its mean mc_{it} is equal to $p_{it} - \log \mu_{it}$ up to a constant $\mathbb{E}[\log \mu_{it}]$, which we include in the first stage.

¹⁸For general multi-input, non-Cobb–Douglas cases, the cost-minimization first-order condition is not linear in inputs and cannot be inverted analytically. Nevertheless, the relationship between ω_{it} and inputs, price, and markups is well-defined and can be approximated by a polynomial of inputs.

¹⁹As firms are price takers and have a markup of 1, the observed output (after substituting the expression for productivity) under perfect competition reduces to $\tilde{y}_{it} = v_{it} + w_t + \eta_{it} - p_{it}$. The last two terms are orthogonal to inputs, v_{it} , and input price w_t .

As noted by Doraszelski and Jaumandreu (2019, 2021), including markups directly in the regression is infeasible—as the whole point of this exercise is to estimate those markups. Even under imperfect competition, however, we only need to control for the variables that determine demand for input (v_{it}), whether those variables operate through markups or through other channels. Thus, one can run a regression similar to (9) that includes determinants of markups rather than the markups themselves. Importantly, when controlling for these determinants, it suffices to know there is a structural relationship between them and the markups; we do not need to know the exact parameters of that relationship.

One potential control, which we use for the rest of the paper, is market share. We do so because this is consistent with the growing literature in macroeconomics and international trade that builds on Atkeson and Burstein (2008) or Kimball (1995), where markups are determined by market share. When disciplining such models with firm-level markup estimates, inserting market share in a first stage is thus internally consistent. Note that market share is not a perfect control for markup and demand conditions in every case—in many industrial organization models, market share does not control for markups (see, e.g., seminal contributions in empirical IO such as Berry, Levinsohn, and Pakes (1995) or Foster, Haltiwanger, and Syverson (2008)).

In summary, the production approach can still be used to estimate markups when output is observed with error. Measurement error affects only estimation of the output elasticity, not the approach in general. One strategy is to purge error in the first stage of a two-stage procedure. Alternatively, the Blundell and Bond (2000) estimator can be used as before, at the expense of greater standard errors.

2.4. Theoretical Extensions

Finally, we use the framework to cover two extensions. Readers initially interested in the main argument may opt to skip these and proceed to Section 3.

2.4.1. Wedges

Macroeconomists often study markups in conjunction with other distortions that create wedges between an input's marginal products and its factor price. Below, we first discuss the extent to which our tools allow any kind of input wedge to be identified, and then discuss how wedges affect the measurement of markups.

Following Hsieh and Klenow (2009), we introduce two wedges in the framework: a non-markup output wedge τ_{it} and an input wedge λ_{it} . In the context of our analytical framework, these affect the firm's perceived profits $\tilde{\pi}_{it}$ as follows: $\tilde{\pi}_{it} = \tau_{it}P_{it}Y_{it} - \lambda_{it}W_tV_{it}$. Both are wedges because they prevent firms from equating marginal revenue products to factor prices. Rather, optimizing firms equate marginal revenue products to the factor price net of wedges: $\frac{\partial P_{it}Y_{it}}{\partial V_{it}} = W_t \frac{\lambda_{it}}{\tau_{it}}$.

If the variable input is subject to the additional input wedge λ_{it} , our tools identify the complete wedge $\lambda_{it}\mu_{it}$ rather than the markup. Inserting the input price distortion into the cost-minimization problem in Section 2.1, the markup equation (1) in the presence of the input price distortion reads

$$\mu_{it}\lambda_{it} = \alpha_{it} \frac{P_{it}Y_{it}}{W_tV_{it}}. \quad (10)$$

Thus, if markups are measured along (1), the estimated markups are the product of the true markup and the input wedge.²⁰ Researchers solely interested in measuring markups should thus, if they observe multiple variable inputs, calculate markups using the input that is the least likely to be subject to input wedges.

Output wedge τ_{it} cannot be estimated using our tools and will not distort the measurement of markups, assuming researchers observe actual revenue $P_{it}Y_{it}$ and not $\tau_{it}P_{it}Y_{it}$. Intuitively, (1) is derived from firms' cost-minimization problem rather than from profit maximization, and revenue does not appear in the former. This is why τ_{it} does not appear in (10), either. Output wedges can distort markup estimation if researchers observe $P_{it}Y_{it}\hat{\tau}_{it}$, where $\hat{\tau}_{it}$ is an output wedge that directly affects measured revenue. The markup estimate is $\mu_{it}\hat{\tau}_{it}\lambda_{it} = \alpha_{it} \frac{P_{it}Y_{it}\hat{\tau}_{it}}{W_tV_{it}}$.

In practice, revenue is usually well-measured and output wedges are implicit, for example, they reflect the fact that firms face financing or regulatory constraints that prevent them from achieving their optimal scale. Output wedges are therefore less likely to obfuscate markup estimation than input wedges are.²¹

Finally, note that wedges on inputs other than V_{it} do not affect the measurement of markups, but these wedges can be estimated using our tools. To see this, consider the case where the firm maximizes profits $\pi_{it} = P_{it}Y_{it} - W_tV_{it} - \lambda_{it}^K P_t^K K_{it}$, where K_{it} is an additional input, with factor price P_t^K , which is subject to the input wedge λ_{it}^K . The new wedge does not alter markup estimates because equation (1) is derived from cost minimization with respect to V_{it} , in which terms with K_{it} drop. One could, however, use the markup equation to identify λ_{it}^K if it was the object of interest. After using the production function estimation tools to identify output elasticity α_{it}^K , the wedge could be calculated as $\lambda_{it}^K = \frac{\alpha_{it}^K P_t^K K_{it}}{\alpha_{it} W_t V_{it}}$.

2.4.2. Quality Differences

An alternative concern, even with quantity data, is that firms within an industry may produce goods of varying quality. If these firms do not share a common production function, one must estimate separate production functions by (often unobserved) quality. A more common assumption, however, is that quality raises demand, as consumers derive greater utility from better products.²² We consider the case where quality affects only demand, and where it also impacts costs.

Quality as a Demand Shifter. If quality affects demand but not marginal costs, as in endogenous growth models, quality differences have limited effects on production function and markup estimation. To see this, assume that demand is a function of price p_{it} and quality q_{it} , that is, $y_{it} = d(p_{it}, q_{it})$. For ease of exposition, keep the simple production function $y_{it} = \alpha v_{it} + \omega_{it}$, with i.i.d. productivity ω_{it} . If output is measured without error, the standard IV-GMM estimator that relies on moment condition $\mathbb{E}[\hat{\omega}_{it}v_{it-1}] = 0$, where $\hat{\omega}_{it} = y_{it} - \hat{\alpha}v_{it}$, still delivers a consistent estimate of α . This is obvious, as the proof for

²⁰Morlacco (2019) uses this property to identify markdowns in the input market for French manufacturing firms. Hashemi, Kirov, and Traina (2022) note that if firms have constant markups and when output elasticities are replaced by revenue elasticities, (1) measures input wedges rather than markups.

²¹It is tempting to think that (1) measures the product of all output and input wedges as the marginal product is $\alpha Y_{it}/V_{it}$; hence, the wedge between the marginal product and W_t is $\alpha Y_{it}/(W_t V_{it})$. That wedge does not equal (1), however, as (1) has revenue rather than output in the numerator.

²²Endogenous growth models, for instance, posit that innovation improves product quality, enhancing long-run welfare (e.g., Grossman and Helpman (1991), ABBK+ (2022)).

consistency of this estimator makes no assumptions on the demand system that firms face (see Supplemental Appendix A.1).

Turning to the case where output is measured with an error η_{it} , quality could affect identification of the production function, but only if researchers purge the error using the two-stage procedure. If researchers deploy an estimator that does not purge errors (i.e., [Blundell and Bond \(2000\)](#)), standard moment conditions still deliver a consistent estimate of α . This is unchanged from the previous results as, again, the consistency derivations do not rely on assumptions on demand.

Quality differences may be more problematic if researchers purge η_{it} . Doing so requires that firms' demand for the variable input is an invertible function of productivity, $v_{it} = v(\omega_{it}, \Xi_{it})$. If quality affects firms' variable input demand as part of Ξ_{it} , one must control for quality to assure that the function can be inverted, $\omega_{it} = v^{-1}(v_{it}, \Xi_{it})$, to run the purging regression $\tilde{y}_{it} = \alpha v_{it} + v^{-1}(v_{it}, \Xi_{it}) + \eta_{it}$.

In our analytical framework, additional quality controls are only necessary if quality affects firms' markups. This is because $v^{-1}(v_{it}, \Xi_{it}) = (1 - \alpha)v_{it} - p_{it} + \log \mu_{it} + w_{it}$, as we show in Section 2.3 without making assumptions on demand. Markups may depend on quality (e.g., because high-income households demand higher-quality products), and have lower demand elasticities ([Nord \(2023\)](#)). The first-stage regression of observed quantity on v_{it} , price p_{it} , and time-fixed effects for w_{it} , then needs additional controls for quality to cover the markup μ_{it} . If quality is constant over time—as is commonly assumed in demand estimations following [Nevo \(2001\)](#)—adding firm-fixed effects to the first stage suffices to consistently purge η_{it} . In settings where quality's effect on markups is less easily controlled for, it is prudent to use [Blundell and Bond \(2000\)](#)'s approach instead.

Quality That Raises Costs and Demand. As firms within an industry have common parameters of the production function, quality differences that raise the cost of production are akin to lower productivity: higher-quality products require more inputs to produce the same output quantity. Below, we show that when researchers apply standard production function methods, the implicit assumption is that quality-adjusted productivity $\omega_{it} - q_{it}$ has the same stochastic process as ω_{it} .²³ To see this, assume that y_{it} is still a firm's quantity of output, unadjusted for quality. The production function now reads as $y_{it} = \omega_{it} - q_{it} + \alpha v_{it}$.

If output is observed without measurement error, the IV-GMM estimator is consistent if the moment condition that held for productivity in fact holds for the sum of quality and productivity. Formally, the moment condition $\mathbb{E}[\hat{\omega}_{it}v_{it-1}] = 0$ will now have $\hat{\omega}_{it} = \omega_{it} - q_{it} - (\hat{\alpha} - \alpha)v_{it}$. The solution to the estimator is given by

$$\hat{\alpha} = \alpha - \frac{\mathbb{E}[q_{it}v_{it-1}]}{\mathbb{E}[v_{it}v_{it-1}]}.$$

The IV-GMM estimator is therefore consistent if quality is orthogonal to the instrument, $\mathbb{E}[q_{it}v_{it-1}] = 0$, which is the assumption that we usually impose on technological productivity, ω_{it} . If one applies the estimator to firms with heterogeneous cost-raising qualities, one thus assumes that the stochastic process of quality-adjusted productivity is the same as

²³There is a separate body of work on how to consistently measure “technological productivity” ω_{it} in the presence of quality differences. Key contributions include [Verhoogen \(2008\)](#), [Kugler and Verhoogen \(2012\)](#), and [De Roux, Eslava, Franco, and Verhoogen \(2021\)](#). [Hahn \(2024\)](#) provides a recent review. An alternative setup, where quality raises both demand and input prices, is studied in [De Loecker, Goldberg, Khandelwal, and Pavcnik \(2016\)](#).

the process assumed for ω_{it} under homogeneous quality. This holds regardless of whether quality affects demand.

If output is measured with error, the IV-GMM estimator can be applied at the expense of greater standard errors. For a two-step estimator, our derivations for cases in which quality is just a demand shifter can readily be applied. There is only a change in labels: where the first stage had to control for productivity, it must now control for quality-adjusted productivity. Inverting variable input demand, we get $\omega_{it} - q_{it} = (1 - \alpha)v_{it} - p_{it} + \log \mu_{it} + w_t$. The purging regression is therefore unchanged from the case in which quality is only a demand shifter.

3. DATA

The remainder of the paper scrutinizes the theoretical predictions from Section 2 with rich quantitative simulations and empirical data. We use data on French manufacturing firms, both to quantify the simulations and for the empirical analysis. We combine FARE (*Fichier Approché des Résultats d'Esane*, Insee (2009–2019b)), which has financial statements, with EAP (*Enquête Annuelle de Production*, Insee (2009–2019a)), which provides product-level revenues and quantities. The sample is from 2009 to 2019. FARE covers the universe of non-financial firms and originates from tax filings. EAP is a product-level survey by the statistical office (INSEE), covering all manufacturing firms with at least 20 employees or 5 million euros in revenue. We detail the data construction in Supplemental Appendix G.

We obtain quantities and prices from EAP, which details a firm's revenue and quantity produced at the level of 10-digit products. For each firm-product, we calculate the ratio of revenue over the quantity of the product sold. We then standardize this unit value by dividing it by the revenue-weighted average price of the 10-digit product across the sample. As some firms produce multiple products, we define a firm's price as the sales-weighted average of standardized prices across the goods that it produces. Quantity is the ratio of revenue over this price.²⁴

We obtain the remainder of the variables from FARE. These are revenue, wage bill, capital, spending on purchased services, and spending on purchased materials (the latter including physical intermediate goods and raw materials). Our sample is the intersection of FARE and EAP. We drop firms with missing, zero, or negative values for any of the variables, and deflate nominal variables using EU-KLEMS deflators.²⁵ Summary statistics are given in Supplemental Appendix Tables G.I and G.II.

4. SIMULATION

We first assess markup estimates in rich quantitative Monte Carlo simulations in a leading macroeconomic model, to scrutinize the estimates in a setting where true markups are known. We use a model that nests and generalizes the analytical framework and that adds an explicit, rich, demand system. We quantify the model using standard parameters and various robustness calibrations in order to assess the quantitative robustness of the analytical conclusions in broad settings.

²⁴As a robustness check, we standardize prices using the revenue-weighted average price at the 8-digit sector level. These firm-level prices have a 0.89 correlation with our baseline prices.

²⁵As in the analytical model, we assume that firms within sectors face identical factor prices. De Loecker, Goldberg, Khandelwal, and Pavcnik (2016) offer a markup estimation method for firms with heterogeneous input prices.

The simulated model is based on Atkeson and Burstein (2008), where firms face double-nested CES demand and compete à la Cournot. The profit-maximizing markup for firm i in market h at time t is a function of a firm's market share s_{iht} :

$$\mu_{iht} = \frac{\varepsilon}{\varepsilon - 1} \left(1 - \frac{\sigma}{\varepsilon - 1} s_{iht} \right)^{-1},$$

where ε is the elasticity of substitution within narrow markets, and σ is the elasticity of substitution across markets. Under the assumption that goods are easier to substitute within markets than across markets (e.g., because goods within a market are more similar), this yields that markups rise in firms' market shares.

Firms produce using two inputs, one variable, v_{iht} , and one fixed, k_{iht} whose endowment follows an AR(1) process. Firms differ in the quantity of the fixed input at their disposal and in their productivity ω_{iht} , which is also AR(1). Given the fixed input and the productivity, firms choose the variable input to minimize costs. Firms combine these inputs using a translog production function:

$$y_{iht} = \omega_{iht} + \gamma \alpha v_{iht} + \gamma(1 - \alpha)k_{iht} + \gamma \frac{\alpha(1 - \alpha)}{2} \frac{\phi - 1}{\phi} (v_{iht}^2 + k_{iht}^2 - 2k_{iht}v_{iht}). \quad (11)$$

The translog production function is a second-order approximation of any production function. Function (11) approximates a CES production function with elasticity of substitution ϕ , degree of homogeneity γ , and variable input weight α . It nests the analytical framework's production function if $\alpha = \phi = 1$ and $\gamma \in (0, 1)$.

The economy is subject to two aggregate shocks: aggregate demand D_t , which scales up demand at given prices, and factor price W_t , which affects demand for the variable inputs. Both are exogenous AR(1) processes. Finally, the econometrician observes output $\tilde{y}_{iht}$, which is the sum of y_{iht} and measurement error η_{iht} .

Market share, markups, quantity, and input usage are endogenous and determined in equilibrium. We provide detailed derivations of the production function, the demand system, and equilibrium factor demand in Supplemental Appendix B, which also details how parameters are chosen based on data from Section 3.

We perform 200 Monte Carlo simulations. Each simulation has 1600 firms, the average firm count in two-digit industries in EAP. We divide firms into 180 markets, the level at which firms compete, and simulate the economy for 40 periods.

4.1. Estimation

Turning to the estimation procedure, we assume that researchers correctly estimate a translog production function with an AR(1) process for productivity

$$y_{iht} = \beta_v v_{iht} + \beta_k k_{iht} + \beta_{vv} v_{iht}^2 + \beta_{kk} k_{iht}^2 + \beta_{vk} k_{iht} v_{iht} + \omega_{iht},$$

where ω_{iht} is productivity which follows $\omega_{iht} = \rho \omega_{iht-1} + \xi_{iht}$, while β 's and ρ are to be estimated. We then use the v_{iht} , to estimate markups along equation (1). The output elasticity of v_{iht} is $\frac{\partial y_{iht}}{\partial v_{iht}} = \beta_v + 2\beta_{vv} v_{iht} + \beta_{vk} k_{iht}$, which varies across firms. The estimated production function is consistent with the model's true one (equation (11)). The β 's satisfy the following relations with the true production parameters: $\beta_v = \gamma\alpha$, $\beta_k = \gamma(1 - \alpha)$, $\beta_{vv} = \gamma \frac{\alpha(1 - \alpha)}{2} \frac{\phi - 1}{\phi}$, $\beta_{kk} = \beta_{vv}$, $\beta_{vk} = -2\beta_{vv}$. Importantly, we do not impose these theoretical relations in the estimation.

We estimate the production function and compare resultant markups across four specifications. The baseline specification uses observed output $\tilde{y}_{iht}$ in two stages. First, $\tilde{y}_{iht}$ is purged of measurement error by regressing it on a third-order polynomial of inputs v_{iht} and k_{iht} , time-fixed effects, price, and market share. As explained in Section 2.3, the residual from this regression consistently captures measurement error. Purged output is then the dependent variable in an IV-GMM estimation using past inputs as instruments. Supplemental Appendix C elaborates.

The second specification uses revenue instead of quantity. We run the first stage without price and market share, as researchers with only revenue data would be unable to control for prices. Additionally, this first stage matches the one frequently used in prior work, following Akerberg, Caves, and Frazer (2015) (ACF below).

In the third and fourth specifications, we estimate the production function without a first-stage regression. The third specification uses quantity and follows Blundell and Bond (2000), which is an application of Arellano and Bond (1991) and Blundell and Bond (1998). Essentially, this specification uses lagged first-differences as instruments for equations in levels, in addition to the usual lagged levels as instruments for equations in first-differences. This is the specification recommended by Doraszelski and Jaumandreu (2019, 2021). In the fourth specification, we run the same estimation using revenue as the dependent variable.

4.2. Results

We assess the quality of quantity- and revenue-based markup estimates in two steps. Section 4.2.1 provides a detailed assessment of the production function and markup estimates for a preferred calibration based on the micro data. Section 4.2.2 shows that the assessment is robust to significant parameter changes.

4.2.1. Main Calibration

In the main calibration, we quantify the model using parameter values that are either standard in the literature or that directly come from the French data described in Section 3. Full details on the calibration are provided in Supplemental Appendix B1.

Production Function Estimates. We first estimate the production function using both the two-stage baseline specification with quantity data, and the two-stage “ACF” specification with revenue data. Table I presents the estimated production function, showing averages across the 200 Monte Carlo simulations. The baseline specification identifies the

TABLE I
ESTIMATED PRODUCTION FUNCTION PARAMETERS.

	$\beta_v = \alpha\gamma$	$\beta_k = (1 - \alpha)\gamma$	$\beta_{vv} = \gamma \frac{\alpha(1-\alpha)}{2} \frac{\phi-1}{\phi}$	$\beta_{kk} = \beta_{vv}$	$\beta_{vk} = -2\beta_{vv}$
True value	0.32	0.48	0.009	0.009	−0.017
Quantity (Baseline)	0.32 (0.002)	0.48 (0.001)	0.009 (0.0005)	0.009 (0.0006)	−0.018 (0.001)
Revenue (ACF)	0.29 (0.003)	0.31 (0.003)	0.005 (0.002)	−0.001 (0.004)	−0.016 (0.005)

Note: “True value” coefficients are directly calculated from the calibrated parameters (α, γ, ϕ) . “Baseline”: A two-stage IV-GMM on observed quantity. “ACF”: A two-stage IV-GMM on revenue. Production function coefficients are averaged across 200 Monte Carlo simulations. Standard deviations across the Monte Carlo simulations are given in parentheses.

production function with precision, with all coefficients equal to or within a standard deviation of their true value. Estimates are also consistent across simulations, as indicated by low standard deviations. The average elasticity of output to the variable input, which determines the average estimate of the markup, equals 0.294, and is close to the true average of 0.295.

The “ACF” specification does not identify the production function coefficients, as expected when using revenue instead of quantity data. The average estimate for β_v of 0.29 is slightly lower than the true value of 0.32. The other coefficients affecting the elasticity of output to the variable input, β_{vv} and β_{vk} , are both closer to zero than their true value, although the higher standard errors show that these coefficients are estimated with considerable variability. More strikingly, the coefficient for the fixed input, β_k , falls from 0.48 to 0.31, and the one for the fixed input squared, β_{kk} , flips sign. The latter are not used to compute the output elasticity of v , so they do not affect the average estimate of the markup. The average estimated output elasticity of v is 0.296, again close to the true value of 0.295.

These estimates are consistent with our analytical results. Section 2.2.2 showed that revenue-based estimates of the production function can be biased upward, or downward, or be unaffected, depending on the correlation between prices and inputs. Section 2.2.3 showed that downward-sloping demand curves cause a downward bias on the estimated elasticity in the absence of demand shocks. Our simulated firms *are* subject to aggregate demand shocks, which create a positive correlation between input usage and prices under diminishing returns to scale, limiting the impact of the bias coming from downward-sloping demand.

Markup Estimates. We next use these coefficients to estimate markups. The results are summarized in Table II. We calculate the correlations with true (log) markups for each simulation and present the average, with standard deviations in parentheses. The baseline estimates are close to true markups. The correlation is close to 1, and the mean, standard deviation, median, and interquartile range are estimated to within a tenth of a decimal point. The slight deviations between true markups and estimated markups are in line with the modest differences between the true and estimated coefficients in Table I, and may be caused, for example, by the fact that the first-stage regression approximates the implicit relationship between productivity and inputs through a third-order polynomial.

The bottom panel of Table II describes the markup estimates based on revenue data. These are highly informative of true markups, with a correlation of 0.94 between the true markup and the revenue-based “ACF” markups. These results again show that the

TABLE II
OVERVIEW—MARKUP ESTIMATES.

	Correlation With True Markups	Log Markup Moments			
		Mean	St. Dev.	Median	IQR
True markups	1.00	0.23	0.07	0.21	0.09
Quantity (baseline)	1.00 (0.01)	0.22 (0.016)	0.07 (0.001)	0.21 (0.016)	0.09 (0.002)
Revenue (ACF)	0.94 (0.05)	0.23 (0.020)	0.07 (0.011)	0.21 (0.021)	0.08 (0.015)

Note: The first column presents estimates' correlations with true markups. Subsequent columns show moments of the estimated (log) markup distribution. Standard deviations across 200 Monte Carlo simulations are in parentheses. “Baseline”: A two-stage IV-GMM on observed quantity. “ACF”: A two-stage IV-GMM on revenue.

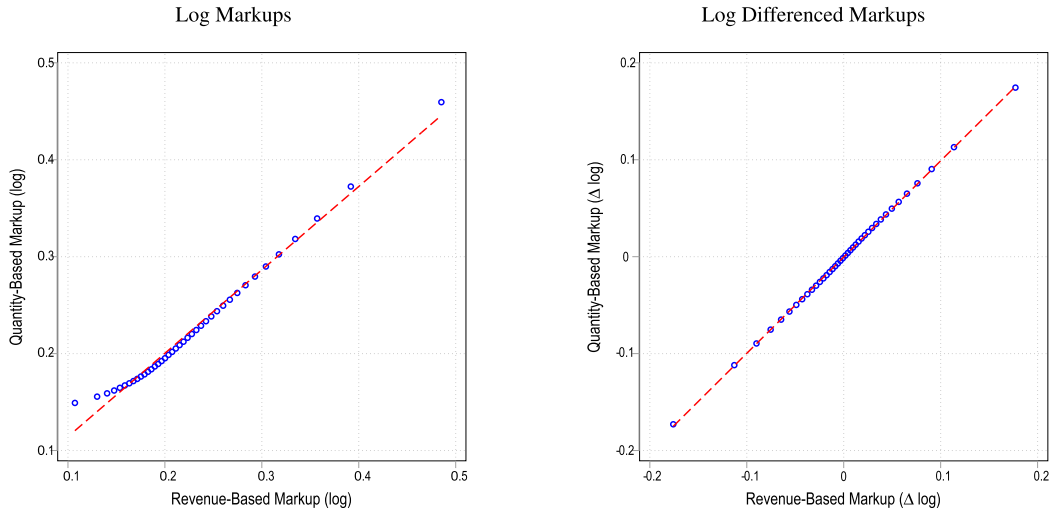


FIGURE 1.—Binned scatter plot for simulated quantity and revenue markups. *Notes:* The figures plot the relationship between quantity-based markups (“baseline”) and revenue-based markups (“ACF”) in simulated data. Log markups are used in panel (a), log differenced markups in panel (b). Linear regression coefficients are 0.85 and 0.99, respectively. The scatters are averages across Monte Carlo simulation.

revenue-based estimates of the production function elasticities are *not* the revenue elasticities of an input. If they had been, log markups should equal 0 and be uninformative of true markups (Section 2.2.1). Rather, the revenue-based elasticities are *biased* estimates of output elasticities of the inputs.

From Section 2.2.4, it follows that a high correlation between true markups and revenue-based estimates requires that prices have low correlations with the production function instruments. Supplemental Appendix Table F.I confirms that correlations are low in the simulations, and that the low correlations are consistent with the data.

Figure 1 provides a graphical illustration of the high correlation between revenue-based markup estimates and true markups. It presents a binned scatter plot between the baseline and “ACF” estimates in log levels (left-hand panel) and in log first-differences (right-hand panel). The plots confirm that quantity-based and revenue-based markups are tightly linked over the entire distribution, especially in first-differences. The regression coefficients are 0.85 and 0.99, respectively, and the linear fit in first-differences resembles a perfect 45-degree line.

Markup Correlations. We next examine how markup estimates correlate with variables from seminal regressions such as markups on the profits rate (operating profit over sales), the material share (ratio of variable-input spending over sales), and market share. For both revenue and quantity-based estimates, we run

$$x_{it} = \chi(\ln \hat{\mu}_{it}) + \varphi_i + \psi_t + \epsilon_{it}, \quad (12)$$

where φ_i and ψ_t denote firm- and time-effects.²⁶ We divide log markup estimates by their standard deviations to ease comparison of rows. The idea is to check whether these re-

²⁶The regression coefficients are qualitatively and quantitatively similar if firm-fixed effects are removed from the estimation equation.

TABLE III
SIMULATED RELATION BETWEEN MARKUP ESTIMATES AND OTHER VARIABLES.

	Profit Rate	Material Share	Market Share
True Markups	0.019 (0.0001)	−0.019 (0.0001)	0.068 (0.0001)
<i>R-Squared</i>	0.901	0.901	0.996
Quantity (Baseline)	0.020 (0.0001)	−0.020 (0.0001)	0.020 (0.0001)
<i>R-Squared</i>	0.899	0.899	0.995
Revenue (ACF)	0.021 (0.0001)	−0.021 (0.0001)	0.075 (0.0002)
<i>R-Squared</i>	0.902	0.902	0.967

Note: OLS coefficients. Explanatory variables are in the column headers. “Baseline”: IV-GMM on quantity. “ACF”: IV-GMM on revenue. Details in Section 4.1. Log markups are normalized to have unit standard deviations so that χ measures the effect of a one-standard-deviation change in markups on x_{it} . Firm-clustered standard errors in parentheses. All specifications include time- and firm-fixed effects. OLS coefficients, standard errors, and R^2 's are averages across the Monte Carlo simulations.

gressions are similar for markups based on revenue and quantity data. If so, this supports prior work using revenue markups in such regressions.

Table III presents the results, confirming that revenue-based markup estimates do well at retrieving the OLS coefficient χ for true markups. Table III shows that if researchers are unaware of biases from the use of revenue data, relationships between markups and other variables remain well-estimated. Market share's R^2 is particularly high, as expected in an [Atkeson and Burstein \(2008\)](#) model.

As we note in Section 2.2.5, there is an alternative approach to avoid biases in regressions such as these. As the output elasticity of v_{it} is pinned down by the interaction of sector-fixed effects and input usage, one could add these as controls to the regression, as long as they are not the object of interest. The remaining markup variation then comes from v_{it} 's revenue share. That is the approach advocated by [Bond, Hashemi, Kaplan, and Zoch \(2021\)](#) and [De Loecker and Warzynski \(2012\)](#).

Measurement Error. Thus far, we have purged output from measurement error through a first-stage regression. There are settings where purging is not feasible, however, for example when the variables that determine firms' variable input demand are unknown or unobserved ([Doraszelski and Jaumandreu \(2019, 2021\)](#)), or when there are quality differences between products (see Section 2.4.2). As explained in Section 2.3, the [Blundell and Bond \(2000\)](#) estimator consistently estimates the production function without purging, at the expense of greater asymptotic variance. We can use the simulations to assess the impact of this higher variance on the markup estimates at various levels of measurement error. In our calibration, 9.5% of the variation in quantity comes from measurement error, consistent with the French data (see Supplemental Appendix B.I). Below, we estimate markups using the [Blundell and Bond](#) estimator with the baseline measurement error, and also subsequently when the error's standard deviation is 2, 4, 8, or 16 times larger.

Figure 2 presents the results. At the baseline measurement error, [Blundell and Bond \(2000\)](#)'s estimator in the left-hand panel performs very well. The average correlation of markup estimates is close to 1, regardless of whether quantity (solid) or revenue (dashed) data are used. For higher levels of measurement error, the two-stage procedures in the right-hand panel perform better. Even when the error's standard deviation is 16 times the baseline level, the quantity estimates have a correlation with true markups of 0.88, and

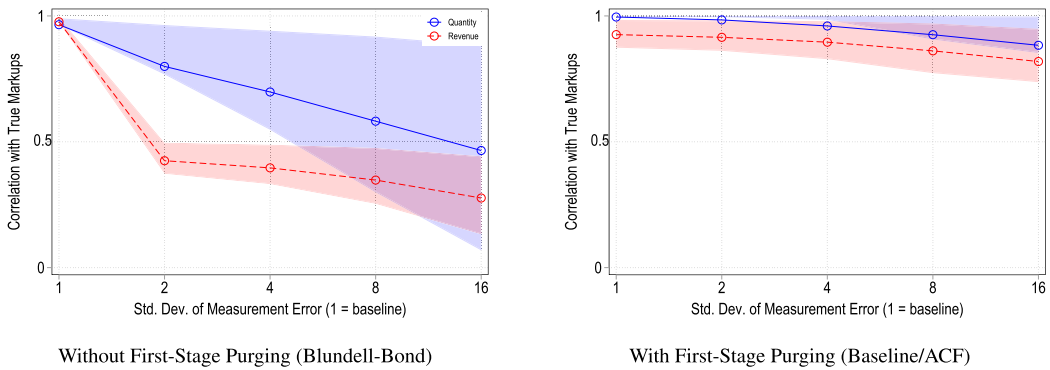


FIGURE 2.—Effect of first-stage purging regression on markup estimates. *Notes:* Figures plot the average correlation between true markups and quantity- (solid) or revenue- (dashed) based markup estimates by standard deviation of measurement error (normalized to 1 for the main calibration). Confidence intervals are the interquantile range. The left-hand panel uses the single-stage [Blundell and Bond \(2000\)](#) estimator; the right-hand panel uses the two-stage baseline and ACF estimator.

the “ACF” revenue estimates have a correlation with true markups of 0.82. Not purging for the error lowers those correlations to 0.47 and 0.28, respectively. It is thus advisable to use a two-stage procedure in settings with significant measurement error. The [Blundell and Bond](#) alternative performs well in settings with low measurement error.

4.2.2. Robustness

Results for the main calibration are robust to substantial changes in the parameters. We perturb all the production function parameters (α , γ , ϕ), and the elasticity of substitution across markets (σ). We vary the variable input’s share in production (α) from 0.4 to 0.9. We raise this parameter because in the baseline (α of 0.4), the average output elasticity is 0.3, on the low end of the sector averages of 0.3 to 0.8 that we find in Section 5 below. We vary returns to scale (γ) from 0.6 to 1.0, symmetrically around the baseline calibration of 0.8. Elasticity of substitution ϕ is 1.10 in the baseline, and we vary this parameter from 0.95 to 1.2.²⁷ Finally, $\sigma > 1$ governs the dispersion of markups. We raise it from the standard value in the literature of 1.1 to 3.1, which reduces true markup dispersion by 78%.

Figure 3 presents the results. The top plots show that the correlation between markups and quantity-based estimates is close to 1 across estimations, while the revenue-based estimates have a correlation with true markups of at least 0.88. This shows that results in Section 4.2.1 are robust to alternative parameterizations.

5. EMPIRICS

We now turn to markup estimation in the EAP-FARE data. True markups are unknown, but we can still assess whether markup estimates based on revenue data are similar to estimates based on quantity data. There are a series of challenges. First, quantities are

²⁷[Oberfield and Raval \(2021\)](#) find a 0.8 elasticity of substitution. We use a higher lower bound, because, for low values, the [Atkeson and Burstein \(2008\)](#) model becomes hard to solve, as marginal costs are highly non-linear in productivity. There is no degradation in the estimators’ performance.

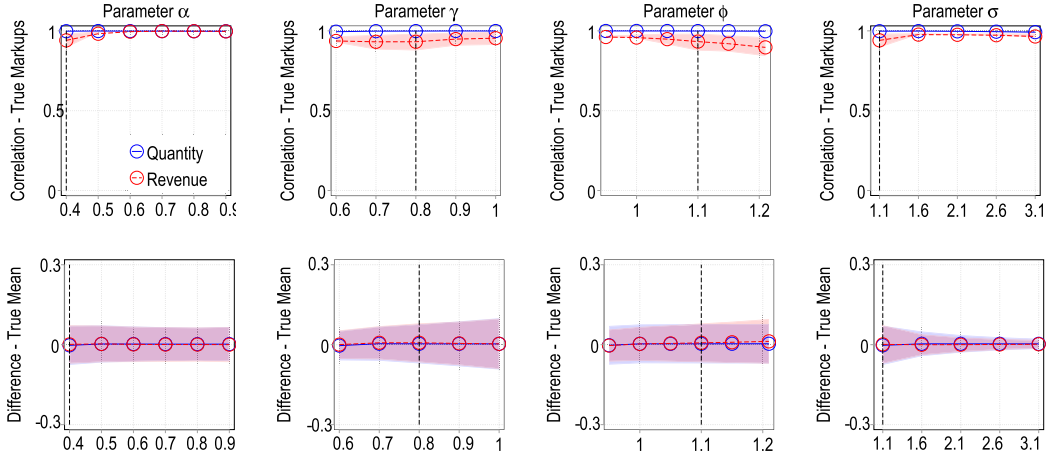


FIGURE 3.—Revenue and quantity markups for alternative parameters. *Notes:* Figures plot quantity-based (Baseline; solid) and revenue-based (ACF; dashed) markup estimates. Vertical lines give baseline values for each column's parameter. Upper figures: correlation between estimates and true markups, interquartile range in shaded areas. Bottom figures: average deviation of estimated markups' mean from true mean; shaded areas give standard deviations of that deviation across Monte Carlo simulations.

approximated by the ratio of revenue over relative prices, as explained in Section 3. Second, firms are unlikely to have identical production functions and, more generally, the translog production function and the AR(1) productivity process may not accurately reflect true production processes. The errors that this causes are specific to each estimation and may differ for quantity and revenue. Despite these complications, we show below that revenue- and quantity-based markups are highly correlated within industries, especially in first-differences, and have reasonably similar variances. We also show that revenue- and quantity-based markups have similar relationships with key variables such as the profit rate, labor, materials, and the market share.

5.1. Production Function Estimation

In contrast to the single sector in the simulations and the analytical framework, the empirical analysis has 18 two-digit manufacturing industries, summarized in Supplemental Appendix Table G.II. We assume that firms only share parameters of the production function and productivity process within industries, and thus perform a separate estimation by industry. Specifically, we assume that log output y_{it} is a translog function of the log of materials m_{it} , the wage bill l_{it} , capital k_{it} , and service inputs o_{it} :

$$y_{it} = \omega_{it} + \beta_m^l m_{it} + \beta_l^l l_{it} + \beta_k^l k_{it} + \beta_o^l o_{it} + \sum_{\{h,j\} \in \{m,l,k,o\}} \beta_{hj}^l h_{it} j_{it}, \quad \forall i \in I, \quad (13)$$

where productivity ω_{it} follows AR(1) process $\omega_{it} = \rho^l \omega_{it-1} + \xi_{it}$ for two-digit sector I . We assume that materials, m_{it} , correspond to variable input v_{it} in Section 2. To estimate markups, we are therefore interested in output elasticity α_{it}^m :

$$\alpha_{it}^m \equiv \partial y_{it} / \partial m_{it} = \beta_m^l + 2\beta_{mm}^l m_{it} + \beta_{mo}^l o_{it} + \beta_{ml}^l l_{it} + \beta_{mk}^l k_{it}.$$

Note that firms within an industry do not have the same output elasticities, α_{it}^m , as the elasticity depends on the level of each input that the firm uses.

TABLE IV
SUMMARY STATISTICS—WITHIN-SECTOR MOMENTS OF THE LOG MARKUP ESTIMATES.

	Mean	St. Dev.	Median*	25th Pct.*	75th Pct.*
Quantity (baseline)	0.366 (0.003)	0.232 (0.003)	−0.011 (0.004)	−0.149 (0.004)	0.137 (0.004)
Revenue (ACF)	0.128 (0.002)	0.163 (0.002)	−0.015 (0.003)	−0.113 (0.003)	0.099 (0.004)

Note: Moments are calculated at the sector level and then averaged across the sectors in the data. Parentheses present bootstrapped standard errors of the moment. “Baseline”: two-stage IV-GMM on quantity. “ACF”: two-stage IV-GMM on revenue. 147,704 obs. *: Medians and percentiles are expressed in deviation of sector averages.

5.2. Results

As in the simulations, we estimate production functions using a two-stage procedure with either quantity (baseline) or revenue (ACF) as the dependent variable.²⁸ Estimated output elasticities average 0.54 across all sectors in the baseline estimation, with a standard deviation of 0.29. Supplemental Appendix Table E.I lists separate averages by two-digit industry. The average output elasticity of materials is 0.40 using revenue, with a standard deviation of 0.14. Average quantity-based elasticities are higher than revenue-based elasticities in 16 of 18 industries. On average, quantity-based elasticities exceed revenue-based elasticities by 38%.

Markup Estimates. We next use these estimated firm-level elasticities to compute markups along equation (1). In the remaining analysis, we focus on the log of markups. To treat for outliers, we trim the bottom and top of the distribution at the 1.5% level for each specification, and perform the analysis on firms for which all estimates fall within the non-trimmed sample, leaving 147,403 observations.²⁹

To make the empirical results comparable to the simulations and the analytical framework, we compare revenue- and quantity-based markups within industries. Table IV presents the estimates’ moments. Each moment is calculated at the sector level and subsequently averaged across industries. The table presents unweighted averages, but results are similar when sectors are weighted by size.

We find an average log markup of 0.37 using quantity data and 0.13 for revenue data. That gap wedge is consistent with the lower estimated output elasticities from revenue data. In contrast to the simulation, we also find a gap between the standard deviation of revenue-based markups and quantity-based markups. On average, quantity-based markups have a standard deviation of 0.23, while revenue-based markups have a standard deviation of 0.16. The medians and percentiles show that the lower dispersion of revenue-based markups is not driven by the tails. In the analytical framework, the standard deviations would be equal, because the production function is Cobb–Douglas. In Table II’s simulation, however, revenue markups had similar standard deviations for sizable deviations from Cobb–Douglas. It is possible that various empirical complications explain the wedge: the production function and productivity process are unlikely to be exactly equal across firms within an industry, and may also not be exactly equal to equation (13). Below, we show that despite these limitations, overall correlations between revenue- and quantity-based markups are high.

²⁸The Supplemental Appendix also provides results for the single-stage Blundell and Bond (2000) estimator.
²⁹The sample reduction is largely because of high variance of the Blundell and Bond (2000) elasticities. These are negative for a non-negligible fraction of firms, for whom log markups are not defined.

TABLE V
CORRELATION BETWEEN QUANTITY- AND REVENUE-BASED LOG MARKUP ESTIMATES.

	Summary of Within-Sector Correlations					Pooled
	Average	St. Dev.	Median	25th Pct.	75th Pct.	Overall
Pearson Correlation	0.61	(0.23)	0.65	0.41	0.78	0.34
Pearson Corr. First Diff.	0.80	(0.16)	0.84	0.70	0.94	0.79
Rank Correlation	0.62	(0.23)	0.68	0.39	0.79	0.43
Rank Corr. First Diff.	0.83	(0.12)	0.84	0.71	0.95	0.84

Note: The table presents correlations between revenue- and quantity-based markup estimates. Correlations are calculated at the two-digit sector in the left-hand panel, and for all observations pooled in the right-hand panel. The summary statistics are averages of the correlations across sectors. 147,704 observations.

Markup Correlations. Table V presents the correlation between revenue- and quantity-based markup estimates in the data. The main panel summarizes sector-level correlations, which are directly comparable to the simulations in Table II. The average simple correlation for log markups is 0.61, while the Spearman rank correlation is 0.62. The high within-sector correlations are visible in the majority of sectors: median correlations exceed average correlations, and consistently exceed 0.65 for levels and 0.84 for first-differenced markups.³⁰ Turning to first-differences, the correlations increase to 0.80 for the simple correlation and 0.83 for the rank correlation. This is close to the 0.94 correlation in the simulations, despite the complications of estimating production functions in the data.

The final column of Table V also presents correlations when all firms are pooled. As expected, combining heterogeneous sectors lowers the correlations: the Pearson correlation is 0.34 while the rank correlation is 0.43. Yet the correlation of growth in markups remains high: 0.80 for the Pearson correlation and 0.84 for rank correlations. Growth in revenue markups thus seems a good predictor of growth in quantity-based markups, even when comparing firms across sectors.

To provide an alternative illustration of the extent to which revenue-based markups successfully predict in quantity-based markups, we next assess whether the linear regression of quantity-based markups on revenue-based markups has a coefficient close to 1. If so, $\hat{\mu}_i^R$ will on average equal quantity-based markup estimates up to a sector effect. We plot results in binned scatter plots in Figure 4, both in levels (left) and first-differences (right). The linear regression coefficients equal 0.89 and 0.92, respectively. Both show an excellent fit between the estimates across values of the revenue-based markups, with the linear fit nearing a 45° line for first-differences. This is quantitatively similar to the corresponding simulation plots (Figure 1), where regression coefficients were 0.85 and 0.99.

Finally, we assess whether relationships between markups and key variables depend on the markup specification, as we did for the simulations in Table III. We regress revenue- or quantity-based markups on either the profit rate (the ratio of operating profits over sales), labor share (the ratio of its wage bill over sales), material cost share (the ratio of materials purchased over sales), and market share.

³⁰Specifications based on the Blundell and Bond (2000) estimator in Supplemental Appendix E are qualitatively similar with, in general, lower positive correlations. This might be due to the fact that prices, and thus quantities, may be subject to considerable error in our data. As we show in Section 4.2, two-stage estimators are more robust in such settings.

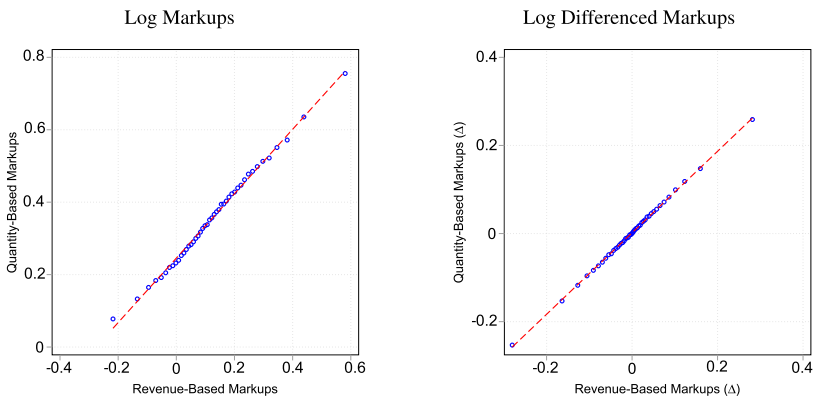


FIGURE 4.—Binned scatter plot: quantity- versus revenue-based markups. *Notes:* The figures plot the binned scatter plot between quantity-based markups (baseline) and revenue-based markups (ACF). Regression coefficients for the linear fit are 0.89 and 0.91, respectively.

Results are presented in Table VI. Rows present regression coefficients for a markup estimate; columns contain the regressor. We pool all observations and normalize the markups to have unit standard deviations. All relationships run in the expected direction: high markup firms are more profitable, have lower labor shares, lower material shares, and greater market shares, irrespective of whether revenue or quantity markups are used. Moreover, the relationships are all significant at the 1% level. Overall, Table VI suggests that estimates of relationships between markups and key variables are qualitatively robust to the use of revenue-based markup estimates. This lends credibility to prior work using revenue-based markups in regressions and further supports our conclusion that these estimates contain useful information about firms’ true markups.

6. AGGREGATE MARKUPS

Finally, we discuss the robustness of trends in aggregate markups. De Loecker, Eeckhout, and Unger (2020) show a significant rise in average firm-level markups since the 1980s for U.S. Compustat firms. This influential result has raised several concerns, mostly about the use of Compustat. We focus on two: (i) revenue is used to proxy for quantity, our main subject of analysis; (ii) they measure trends in simple revenue-weighted averages rather than harmonic or cost-weighted averages, which may not be welfare-relevant

TABLE VI
EMPIRICAL RELATION BETWEEN MARKUP ESTIMATES AND OTHER VARIABLES.

	Profit Rate	Labor Share	Material Share	Market Share
Quantity (baseline)	0.150 (0.002)	−0.050 (0.001)	−0.096 (0.001)	0.070 (0.006)
<i>R-Squared</i>	0.357	0.098	0.312	0.010
Revenue (ACF)	0.111 (0.001)	−0.027 (0.001)	−0.077 (0.001)	0.031 (0.004)
<i>R-Squared</i>	0.502	0.075	0.501	0.007

Note: Each entry provides the OLS estimate using the variable in the column header as the dependent variable and the markup series in the row as the regressor. Firm-clustered standard errors in parentheses. All regressions include time- and firm-fixed effects.

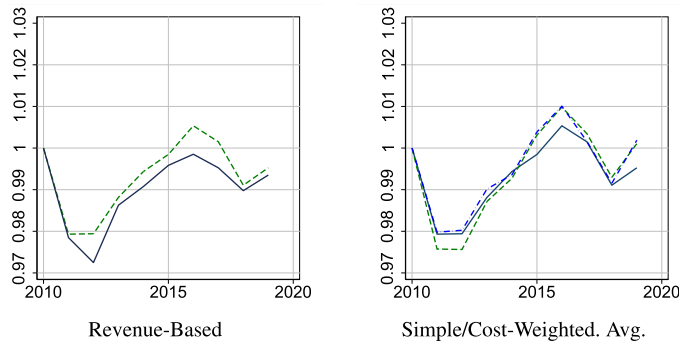


FIGURE 5.—Trend in aggregate markups in France (index = 1 in 2010). *Notes:* Solid lines plot aggregate (harmonic revenue-weighted) markups based on quantity. Dashed lines, from left to right, plot: aggregate markups based on revenue; aggregate markups based on quantity that are simple rather than harmonic weighted. Dash-dotted line in the second plot: aggregate markups based on quantity that are cost weighted. Markup levels are not shown; all are indexed to 1 in 2010.

aggregate markups. We assess the quantitative importance of both critiques for aggregate markups in France below.

We define baseline aggregate markups as the sales-weighted harmonic average of our quantity-based markup estimates, $(\sum_{i \in I_t} s_{it} \mu_{it}^{-1})^{-1}$, where I_t is the set of firms at time t , while s_{it} denotes i 's share in aggregate sales. This is the welfare-relevant measure of the aggregate markups in a broad set of models (see, e.g., Edmond, Midrigan, and Xu (2023), Grassi (2017)). We then deviate from this baseline in two ways: (i) using the revenue-based markups of the “ACF” specification, and (ii) using a simple or cost-weighted average rather than a harmonic sales-weighted average.

Figure 5 shows the resulting time series for aggregate markups, where we investigate trends by normalizing each series to 1 in 2010. The first panel shows that quantity-based and revenue-based aggregate markups follow similar dynamics: they exhibit a decline around the Eurocrisis in 2011 and 2012, followed by an upward trend. Their average level is different, the aggregate revenue-based markups averaging 1.08, while aggregate quantity markups average 1.45. The second panel shows similar movements for the cost-weighted average, the simple sales-weighted average, and our baseline aggregate markup.

We conclude that trends in markups of French firms between 2010 and 2019 are robust to the use of revenue and the type of aggregation, while both of these changes strongly affect the level of the aggregate markup.

7. CONCLUSION

This paper assesses the feasibility of estimating markups from widely used data in the macroeconomics and international trade literature. Practically, we conclude that it depends on the research design whether an analysis can proceed with revenue data. If feasible, production functions for markups should be estimated with quantity data. Given the paucity of such data, we show that revenue data can suffice for researchers interested in dispersion of markups across firms. Revenue data may also be used to estimate trends, if one is willing to assume a constant production function over time. Conversely, in applications that focus on the average level of the markup, revenue data are not appropriate.

To estimate the production function, we recommend using a two-stage estimator in settings where output is observed with significant error. One-stage estimators yield less

precise estimates but are useful when it is difficult to control for the forces determining a firm's demand for inputs or when the quality of goods produced by firms is heterogeneous and unobservable.

There are a number of caveats. Our markup estimates are contingent upon correct specification of the production function. While both our simulation and empirical analysis rely on the translog function, which is a second-order approximation of any production function, approximation error will affect markup estimates. Our approach to measure markups will also capture other input wedges, such as markdowns due to monopsony power, if the variable input is subject to them. A quantification of markups' contribution to overall input wedges, for example with external markup estimates, is an exciting avenue for future research.

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All authors assume responsibility for all aspects of the paper.